

EVOLUTION

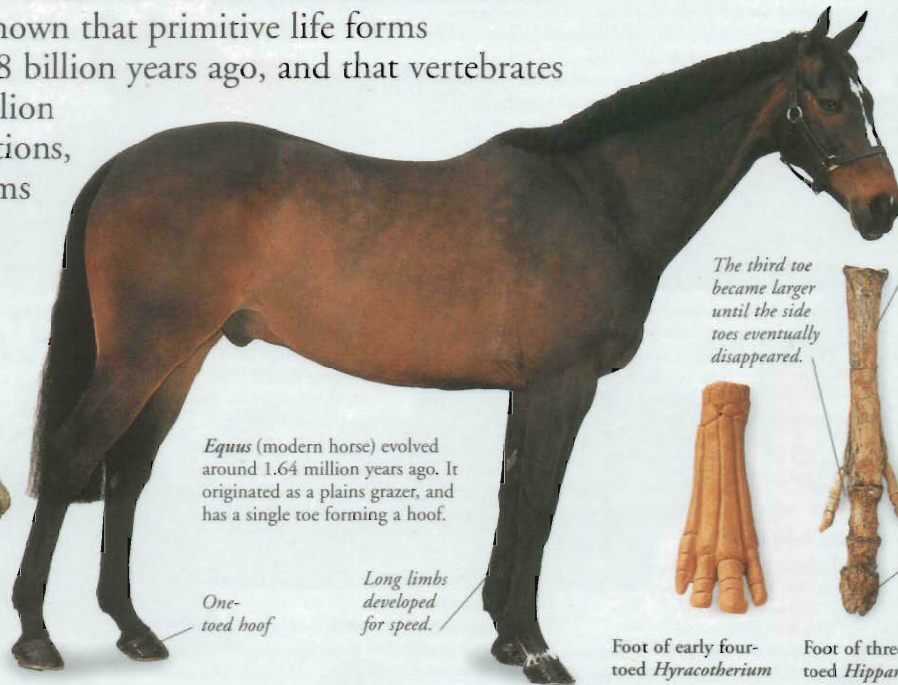


THE TERM EVOLUTION refers to the theory that existing animals and plants have evolved, or developed, through a process of continual change from previous life forms. Some scientists argue that by looking at fossil evidence we can find out more about the past. Fossils have shown that primitive life forms appeared more than 3.8 billion years ago, and that vertebrates existed at least 500 million years ago. Over generations, better-adapted organisms have developed.

Hyracotherium, the first in a long line of horses, appeared approximately 55 million years ago.



Four-toed hoof



Equus (modern horse) evolved around 1.64 million years ago. It originated as a plains grazer, and has a single toe forming a hoof.

One-toed hoof

Long limbs developed for speed.

How a species evolves

An organism may undergo change due to a number of processes, such as natural selection and adaptation, induced by the environment in which it lives. For example, selection may have promoted larger and faster horses adapted for living on open grassy plains. In some cases, a subspecies can change so much that it becomes very different from the rest of its species.

The third toe became larger until the side toes eventually disappeared.

Three-toed Hipparion appeared around 15 million years ago.



Foot of early four-toed *Hyracotherium*



Foot of three-toed *Hipparion*



Third toe

Hoof of modern horse, *Equus*

Natural selection

The theory of natural selection suggests that environmental factors may favour some members of a species over others, enabling the fittest to survive. Experiments with the carnivorous plant, sundew, backed this theory. Plants that were fed meat produced more flowers and seeds than plants which were not.



Plant fed meat produces flowers.

Unfed plant has fewer flowers.

Heredity

Heredity studies how characteristics of certain individuals are passed on from one generation to the next. This process often involves the expression of a parent's dominant gene, which may mask the effect of the recessive gene.

Russian hamsters



Recessive albino gene

Dominant grey gene

Variation

Variation refers to the differences within a species. Certain variations, such as shell colour, may give certain individuals advantages over others. These individuals will have a greater chance of surviving and reproducing.

Banded snails



Shell colours allow snail to adapt to local habitat.

Mutations

Occasionally during reproduction, the process of replicating genetic material (DNA) goes wrong and produces an accidental mutation. Normally, these mutants do not live but some survive as an important source of variation on which selection operates.

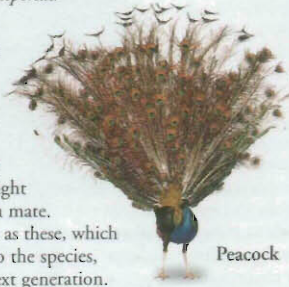


Teasel plant

Fuller's mutant teasel has curved spines, allowing easier seed dispersal.

Sexual selection

Some animals choose their mates by means of sexual selection. A male, may have elaborate features, such as bright feathers, to attract a mate. Characteristics such as these, which may be beneficial to the species, are passed to the next generation.



Peacock

Adaptation

The process of adaptation occurs when an organism evolves in a certain way to make it better suited to its environment. Some people believe that this can lead to a new species, and that an animal or a plant will adapt to its surroundings so that it has a better chance of survival.



Plant-feeding finch



Insect-feeding finch

Darwin's finches

While visiting the Galápagos Islands, English naturalist Charles Darwin (1809–82) noticed the variety of finches living there. Because they resembled finches of mainland South America, he argued that each finch species had evolved a differently shaped beak to meet the needs of its particular diet.



Monarch butterfly



Viceroy butterfly

Mimicry in butterflies

Mimicry is a form of adaptation where one species has developed a resemblance to another as a means of protection. Birds find the monarch butterfly distasteful, but also avoid the similarly coloured viceroy, even though it is palatable.